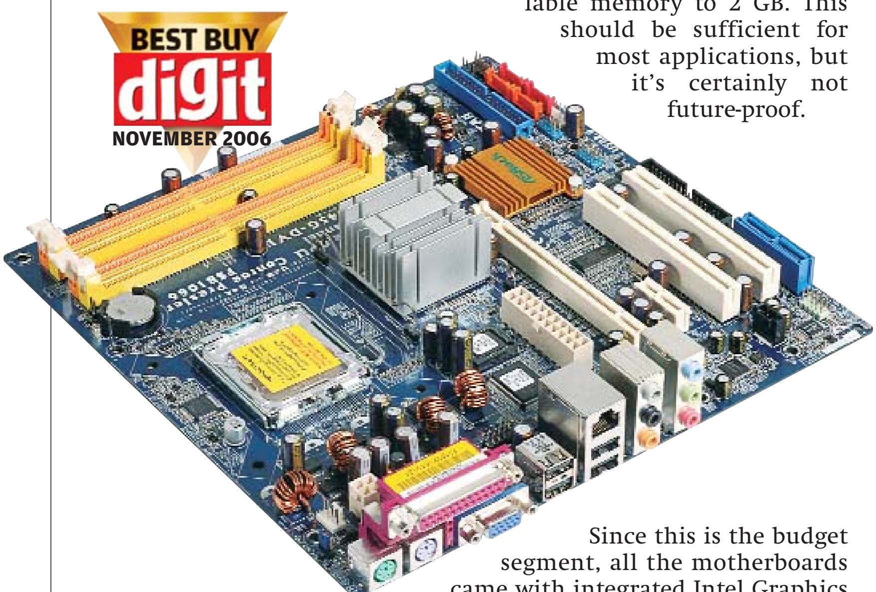
fight it out for top spot.

Designed to support Pentium 4 EE/HT and Pentium D CPUs, this chipset is mainly targeted at budget consumers. With later revisions providing Core 2 Duo support, this category can be considered the poor man's tool to fulfil the Core 2 Duo dream. The ASRock ConRoe 945G-DVI, Gigabyte GA-8I945GZME-RH, and the Gigabyte GA-945GM-S2 are some of the motherboards which, despite their older chipsets, support Core 2 Duos.

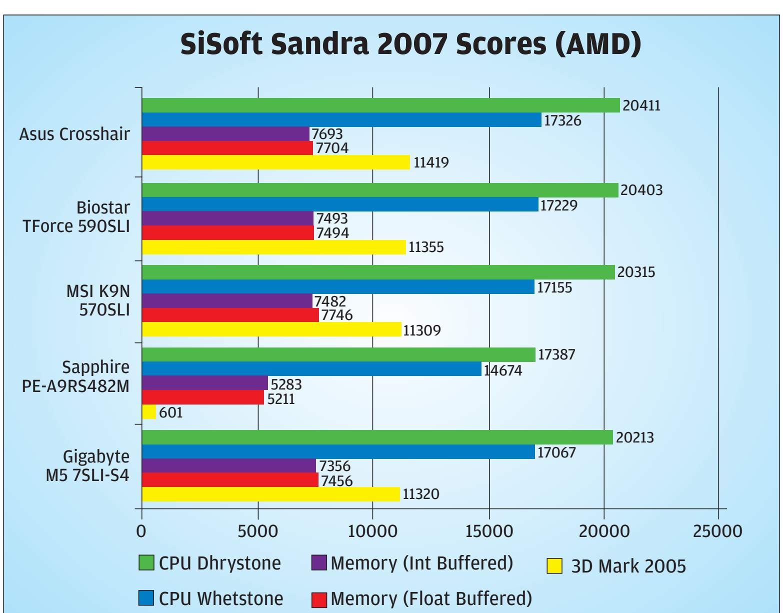
All the 945GZ based chipsets have only two memory slots, limiting the maximum installable memory to 2 GB. This should be sufficient for most applications, but it's certainly not future-proof.



ASRock Conroe 945G-DVI
A David to the G965 Goliaths

Since this is the budget segment, all the motherboards came with integrated Intel Graphics Media Accelerator 950. While the 945G chipset-based motherboards allow for an upgrade via a PCI-Express slot, the 945GZ limits you to the integrated graphics.

All the motherboards have a maximum of four SATA ports. While most of these boards have only one IDE port, the MSI 945G Platinum offers three. It was also the only board with RAID support.



The MSI 945GZM3 has placed the Northbridge heatsink very close to the processor heatsink. Gigabyte's GA-945GM-S2 is another board with a bad layout—the CPU fan power socket is located bang in the middle of the RAM slots. This can be very inconvenient while removing the RAM or the CPU fan power cable.

Since these motherboards fall in the budget segment, it doesn't necessarily mean that they don't come with a good bundle. MSI 945GZM3 comes bundled with Norton Internet Security 2005; ASRock has packed its ConRoe 945G-DVI with McAfee virus scan.

Performance

Gaming

These motherboards are not supposed to be used for high-end gaming, so they have very minimal gaming performance compared to their cousins in the other two categories. Since all motherboards featured integrated video, you may not be able to play all the latest games, but classics like *Quake III* will still rock using onboard graphics.

Synthetic Benchmarks

ASRock's Conroe 945G-DVI, topped the test with a score of 771 units in 3DMark 05. It had a clear advantage due to the Core 2 Duo X6800 it was running. The advantage due to the more powerful CPU was clearly evident as the Pentium 4 powered ECS 945G-M3 returned a score of only 636 units.

The PCMark 2005 test also shows the disparity a CPU can create among similar motherboards. While the 945G-based ECS 945G-M3 could only manage an overall score of 3019, the ASRock Conroe 945G-DVI gave around 60 per cent higher scores, with an overall score of 4929 units. Comparing these scores to those of a G965-based board like the Gigabyte GA-965G-DS3 really shocked us. The 945G chipset-based ASRock Conroe 945G-DVI did better than GA-965G-DS3's score of 4073 units. This clearly shows that ASRock has done a good job with the Conroe 945G-DVI, optimising it for the Core 2 Duos.

SiSoft Sandra 2007 Engineer's edition CPU scores were fluctuating due to the difference between the type of CPU used. The difference between Pentium 4 and Core 2 Duo was huge—especially the CPU multimedia scores.

Real-World Benchmarks

The Gigabyte GA-945GM-S2 and ASRock Conroe 945G-DVI took practically the same time to finish our video encoding test. With 84 seconds, the Gigabyte GA-945GM-S2 was definitely slower than the P965-based motherboards, but faster than the expensive G965 motherboards. The MSI 945GZM3 was the slowest, taking 148 seconds.

Our Conclusion

The ASRock Conroe 945G-DVI led the way in most of the tests due to its support for the more powerful Core 2 Duo CPU, its older chipset notwithstanding. It also did better than some G965 chipset-based motherboards. So if you're looking for a Core 2 Duo board on a budget,